

STAT 482

Capstone in Analytics
for Business

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UAEU · College of Business
and Economics

Modeling the Impact of Environmental Factors on Ecosystem Stress

Using Multiple Regression

Case Study 5 — Forest Fires Dataset (UCI ML Repository)

Dataset at a Glance: Forest Fires (N = 517)

517

fire events recorded

Observations**8**

environmental variables

Predictors**0**

complete dataset

Missing Values**12.85**

ha burned area

Response (mean)

Variable	Min	Median	Mean	Max
temp (°C)	2.20	19.30	18.89	33.30
RH (%)	15.00	42.00	44.29	100.00
wind (km/h)	0.40	4.00	4.02	9.40
DMC	1.10	108.3	110.9	291.3
DC	7.90	664.2	547.9	860.6
area (ha)	0.00	0.52	12.85	1090.84

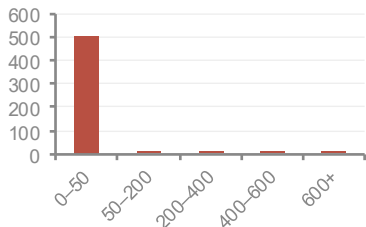
Response Transformation & Predictor Correlations

Response Variable: Burned Area

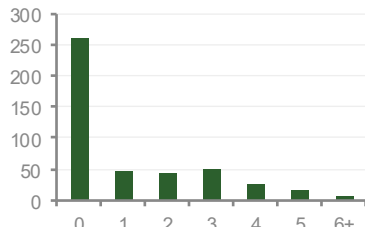
Original area: Heavily right-skewed. Mean = 12.85 ha, Median = 0.52 ha, Max = 1,090.84 ha. Over 50% of fires burned zero measurable area.

Transformation applied: $\log(\text{area} + 1)$ — normalises distribution, compresses extreme values, satisfies OLS assumption of near-normality.

Original area



$\log(\text{area} + 1)$



Key Correlations with log_area

Predictor	r with log_area	Notable inter-correlation
DMC	0.07	DMC-DC: 0.68
temp	0.05	temp-RH: -0.53
wind	0.07	—
RH	-0.05	RH-temp: -0.53
FFMC	0.05	FFMC-ISI: 0.53
DC	0.07	DC-DMC: 0.68
rain	0.02	—

Full OLS Regression Model Results

$$\log(\text{area} + 1) = 0.222 + 0.008 \cdot \text{FFMC} + 0.001 \cdot \text{DMC} + 0.0003 \cdot \text{DC} - 0.024 \cdot \text{ISI} + 0.002 \cdot \text{temp} - 0.005 \cdot \text{RH} + 0.076 \cdot \text{wind}^* + 0.097 \cdot \text{rain}$$

Predictor	Estimate	Std. Error	t-value	p-value	Sig.
(Intercept)	0.2224	1.3604	0.163	0.870	
FFMC	0.0077	0.0145	0.532	0.595	
DMC	0.0012	0.0015	0.814	0.416	
DC	0.0003	0.0004	0.767	0.444	
ISI	-0.0239	0.0169	-1.415	0.158	
temp	0.0025	0.0173	0.143	0.887	
RH	-0.0052	0.0052	-0.997	0.319	
wind	0.0758	0.0366	2.069	0.039	*
rain	0.0965	0.2121	0.455	0.649	

* $p < 0.05$ Wind is the only significant predictor

Model Fit Summary

R^2 0.0199

Adjusted R^2 0.0044

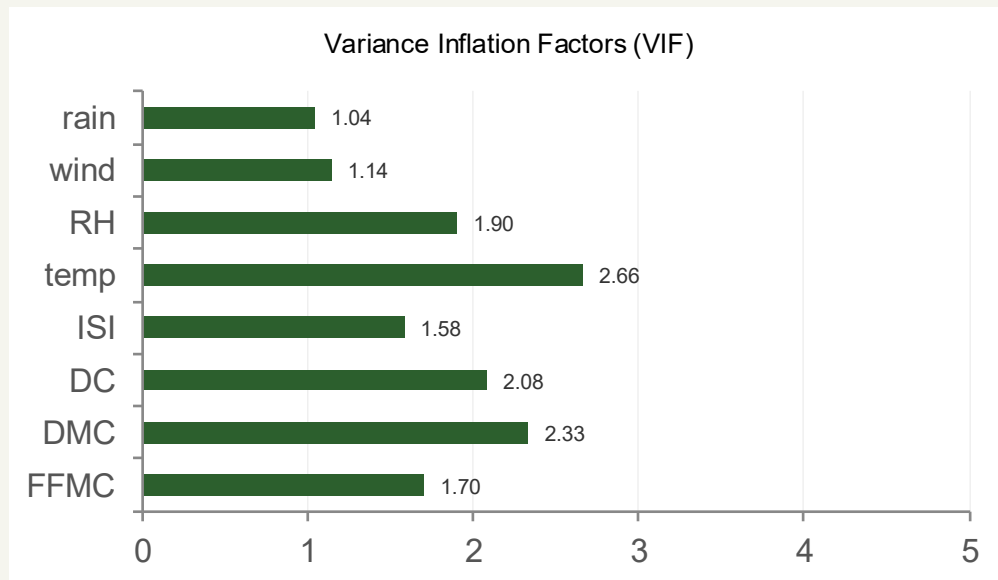
F-stat 1.288

F p-value 0.247

Residual SE 1.395

df (residual) 508

Multicollinearity & Model Assumption Tests



Threshold = 5 ↑ Concern

Assumption Tests

PASS

Breusch-Pagan Test

BP=11.27, $p=0.187$ · No significant heteroscedasticity

FAIL

Shapiro-Wilk Test

W=0.854, $p<2.2e-16$ · Residuals non-normal (heavy right tail)

PASS

VIF (max)

temp = 2.66 · All predictors VIF < 5

WARN

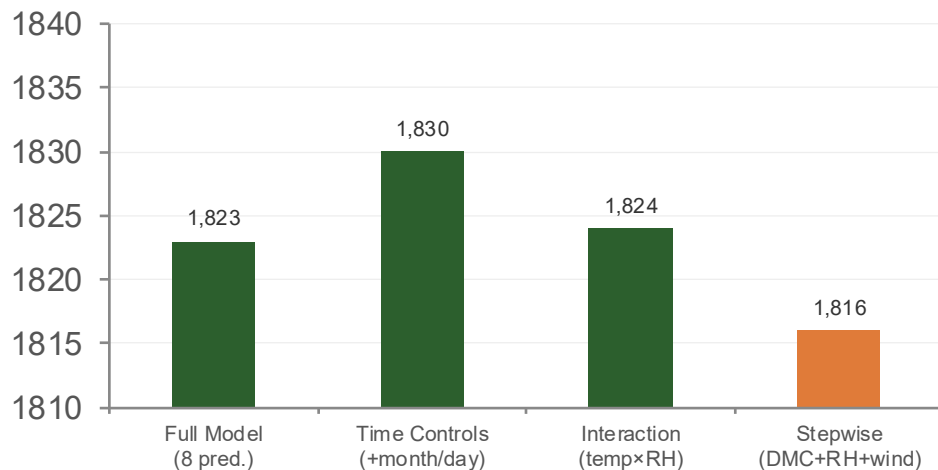
Residuals vs Fitted

Visual: fan-shaped · Some heteroscedastic tendency visible

Key takeaway: No multicollinearity issue (all VIF < 5). Homoscedasticity holds formally. Non-normality of residuals noted — standard errors remain asymptotically valid at $N=517$.

Model Comparison & Stepwise Regression

AIC by Model (lower = better)



Stepwise Model (Best AIC = 1815.52)

$$\log(\text{area}+1) = 0.913 + 0.00175 \cdot \text{DMC} - 0.00558 \cdot \text{RH} + 0.0624 \cdot \text{wind}$$

Predictor	β	p-value
(Intercept)	0.9129	< 0.001 ***
DMC	0.0018	0.070 .
RH	-0.0056	0.140
wind	0.0624	0.071 .

Model Comparison Summary

Model	Adj. R ²	AIC	BIC	F p-value	Verdict
Full (8 pred.)	0.004	1823	1865	0.247	Weak
Time Controls	0.021	1830	1945	0.078	Moderate
Interaction	0.004	1824	1871	0.287	Weak
★ Stepwise (DMC+RH+wind)	0.008	1816	1837	0.061	Best AIC/BIC

Model Validation: Train/Test Split

Data Split

Training set: 80% → 416 observations

Test set: 20% → 101 observations

Method: Stratified (caret)

1.71

log(area+1) scale

RMSE (test set)

1.30

log(area+1) scale

MAE (test set)

Training Model — Key Coefficients

Predictor	β	p-value
FFMC	0.00455	0.759
DMC	0.00050	0.755
DC	0.00060	0.110
ISI	-0.02989	0.088
temp	0.00623	0.739
RH	-0.00298	0.597
wind	0.08286	0.037 *
rain	-0.89116	0.237

Validation Interpretation

RMSE = 1.71 on log scale

Corresponds to $\sim \pm e^{1.71} \approx \pm 5.5$ ha prediction error on original scale.

MAE = 1.30 on log scale

Average absolute error $\approx e^{1.30} \approx 3.7$ ha — substantial uncertainty.

Wind remains significant ($p=0.037$) in training set, consistent with full model.

Low R^2 reflects genuine limit of meteorological-only models for fire extent prediction.

Conclusions & Managerial Implications

KEY FINDINGS

- 01 Wind speed is the sole statistically significant predictor across all models ($\beta=0.076$, $p=0.039$).
- 02 Stepwise selection retains DMC, RH, and wind as the best parsimonious model (AIC=1815, BIC=1837).
- 03 All VIF values < 5 — no multicollinearity concern.
- 04 Homoscedasticity holds (BP $p=0.187$); residuals are non-normal (SW $p<2.2e-16$).
- 05 Test set RMSE=1.71, MAE=1.30 — substantial prediction uncertainty remains.

MANAGERIAL IMPLICATIONS

Wind-based alert systems

Automate fire-risk warnings when wind > 5–6 km/h during drought periods.

DMC-guided prescribed burns

Schedule controlled burns in winter when DMC is low to reduce deep fuel load.

July resource allocation

Model shows elevated risk in July — concentrate suppression resources mid-summer.

Integrate spatial data

Add GIS/topographic layers to surpass the 2% R^2 ceiling of meteorological-only models.